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Sovereign Risk Dynamics in the Eurozone

Modeling and Forecasting the BTP-Bund Spread (2021-2026)

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Contents

1 Introduction **1**

2 Literature Review **3**

3 Data and Methodology **5**

3.1 Data source and Dataset evolution 5

3.2 Methodology 6

3.2.1 Standard Preprocessing Protocol 6

4 ARMA Model **8**

4.1 Sprecification and Estimation 8

4.1.1 Preliminary Stationarity Analysis 8

4.1.2 Formal Stationarity Tests 9

4.1.3 Differencing and Stationarity Verification 9

4.1.4 Autocorrelation Analysis of the Differenced Series 10

4.1.5 Preliminary Tests Confirming Structure 10

4.2 ARIMA Estimation 11

4.2.1 ARIMA Residual Diagnostics 12

4.3 GARCH Model: Specification and Selection 12

4.3.1 Motivation for GARCH 12

4.3.2 Systematic Search for Optimal Model (Grid Search) 12

4.3.3 Positivity and Stationarity Conditions 13

4.3.4 Full GARCH Diagnostics 14

4.3.5 Rolling Validation Strategy 14

5 ARMA-GARCH Validation **15**

5.1 Out-of-Sample Validation with Rolling Window 15

5.2 Distribution Evaluation and Interval Coverage 15

6 VAR Model **17**

6.1 Data Setup 17

6.2 Methodology 18

6.2.1 MQ Test . . . 19
6.2.2 Distribution . . . 20
6.2.3 Granger Test . . . 20
6.2.4 Impulse Response Function . . . 22
6.2.5 FEVD . . . 22
6.2.6 ARCH-LM Test . . . 23

7 DCC Model, Validation, and Forecasting 24
7.1 Introduction . . . 24
7.2 Motivation: Conditional Heteroskedasticity in VAR Residuals . . . 24
7.3 DCC Model Estimation . . . 25
7.3.1 Distributional Choice . . . 25
7.3.2 DCC Parameter Estimates . . . 25
7.3.3 Dynamic Conditional Correlations . . . 27
7.4 Model Diagnostics . . . 28
7.5 Model Validation: Rolling-Window Forecasting . . . 29
7.5.1 Forecasting Methodology . . . 29
7.5.2 Forecast Performance . . . 29
7.5.3 Benchmark Comparison: Random Walk . . . 30
7.6 Discussion and Conclusions . . . 31

Abstract

This study analyzes the dynamics and short-term forecasting of the BTP-Bund spread, a key indicator of sovereign risk in the Eurozone, over the period from April 2021 to April 2026. The report presents an integrated modeling framework that combines a univariate ARMA-GARCH analysis to validate the marginal statistical properties of the series, with a multivariate VAR-DCC-GARCH extension to capture dynamic interdependencies with cross-market variables (VStoxx, 3-month Euribor, 3-month Overnight Index Swap, and the 10Y sovereign spreads of Spain and Ireland with respect to Germany). In line with the literature, the spread is modeled in first differences to ensure stationarity while preserving its economic interpretation in basis points. The empirical results confirm that the differenced series follows a White Noise process in the mean, making point forecasts comparable to a Random Walk. However, the GARCH specification with a Student's t-distribution provides dynamic and well-calibrated confidence intervals, while the VAR-DCC-GARCH extension allows for the explicit modeling of shock transmission and time-varying conditional correlations. Overall, the document provides a coherent methodological framework that combines marginal validation with systemic analysis, offering a robust tool for sovereign risk monitoring and short-term portfolio risk management.

Chapter 1

Introduction

This study examines the dynamics and forecasting of the BTP-Bund spread by adopting an evolutionary methodological approach that reflects the increasing complexity of financial markets. The objective is not to identify a static economic structure, but rather to conduct a comparative evaluation of univariate and multivariate models within an iterative and out-of-sample forecasting framework. The research process developed through iterative stages of refinement. In the initial phases, weekly-frequency analysis (2010–2026) made it possible to identify long-term cycles. However, the framework was subsequently transitioned to a daily frequency since daily observations allow the model to capture immediate market reactions and sudden shocks without being contaminated or biased by structural shifts from past economic regimes. Spectral analysis highlighted the presence of high-frequency noise, justifying the transition to daily data and the reduction of the sample window to the 2021–2026 period, characterized by greater macroeconomic turbulence. At the same time, the model specification was progressively enriched. The analysis evolved from simple ARMA models to ARMA-GARCH frameworks in order to manage the conditional heteroskedasticity typical of sovereign yield differentials. After testing numerous variables,The definitive VAR-DCC-GARCH model includes the BTP-Bund 10Y spread, 3-month Euribor, 3-month OIS, Spanish sovereign spread (ES), and Irish sovereign spread (IE).

The structure of the report is designed as a unified analytical pathway, divided into two complementary macro-phases. The first phase is dedicated to the specification, validation, and forecasting of the univariate ARMA-GARCH model, which provides the necessary diagnostic foundation to ensure stationarity, correct residual distribution, and absence of seasonality. The second phase extends the analysis to the multivariate VAR-DCC-GARCH system, simultaneously modeling conditional mean dynamics and dynamic conditional correlations between the sovereign spread and a basket of financial variables (equity volatility, money market rates, and cross-country spreads). This integrated approach, developed within the same document, makes it possible to assess whether the inclusion of cross-market dependencies improves forecasting performance and risk calibration relative

to purely univariate specifications.

The document is organized to guide the reader through a coherent methodological flow. Sections 4 and 5 present the methodology and results of the univariate ARMA-GARCH module, including preprocessing, order identification, rolling estimation, residual diagnostics, and out-of-sample validation. Subsequent sections extend the framework to the VAR-DCC-GARCH model, presenting the multivariate specification, estimation of dynamic conditional correlations, system stability tests, and comparative evaluation of forecasting performance. This integrated structure allows the contribution of each component (univariate mean, conditional volatility, and cross-market correlations) to be isolated and its joint impact on forecasting accuracy and risk calibration to be assessed.

Chapter 2

Literature Review

The literature on sovereign spreads highlights that these spreads do not solely reflect the credit risk of individual countries, but rather result from a complex combination of fundamental factors, market conditions, and dynamic interactions across financial markets. A first key contribution is provided by Codogno, Favero, and Missale (2003), who argue that sovereign spreads primarily represent a signal of fiscal risk. From this perspective, the bond market performs a disciplinary function toward governments by exerting pressure on fiscal fundamentals and public debt sustainability. Subsequently, Favero et al. (2010) propose a more articulated framework in which spreads depend simultaneously on credit risk, aggregate risk, and market liquidity, emphasizing particularly the interactions among these elements. In particular, liquidity is relevant during periods of stability, but tends to lose explanatory power during crises, when systemic risk becomes dominant. In this direction, Beber et al. (2009) introduce the concepts of flight-to-quality and flight-to-liquidity, showing that during turbulent periods investors tend to shift toward safer and more liquid assets, thereby amplifying spread movements. De Santis (2012) further broadens the framework by showing that sovereign spreads are influenced not only by fiscal and liquidity factors, but also by regional aggregate risk, country-specific risk, and contagion effects across countries. During the European sovereign debt crisis, these interdependencies became particularly significant, highlighting the systemic nature of sovereign risk. Mink and De Haan (2012), analyzing the Greek crisis, further demonstrate that spread dynamics are strongly influenced by expectations regarding interventions by European institutions, thereby introducing an institutional and political dimension to contagion. Finally, Calice et al. (2011) emphasize the interconnection between sovereign bond markets and CDS markets, showing that liquidity and pricing conditions in one market are transmitted to the other, confirming the integrated nature of the financial system. From an econometric perspective, the literature shows that financial time series, including sovereign spreads, exhibit strong persistence and volatility clustering. To model these characteristics, Engle (1982) introduced ARCH models, later generalized by Bollerslev (1986) into GARCH models, which allow for a more parsimonious representation of conditional volatility

dynamics. However, these models are inherently univariate and therefore unable to capture interactions across markets. For this reason, multivariate models have been developed, including Engle's (2002) Dynamic Conditional Correlation (DCC) framework, which makes it possible to model time-varying correlations among multiple financial series. Overall, the literature suggests three main conclusions: (i) sovereign spreads depend on a combination of credit risk, liquidity, and macro-financial factors; (ii) these determinants interact nonlinearly and become particularly relevant during periods of crisis; and (iii) temporal dynamics and market interconnections require econometric tools capable of simultaneously modeling volatility and dynamic correlations. However, the question of how to empirically evaluate the relative contribution of these factors within an out-of-sample forecasting framework remains open, and this constitutes the core focus of the present study.

Chapter 3

Data and Methodology

3.1 Data source and Dataset evolution

The data were sourced from Refinitiv. The construction of the dataset followed a progressive refinement process guided by exploratory analysis.

- Data Frequency:

  In the initial stages of the study, weekly data (from 2010 or 2016 onward) were used to analyze long-term trends. To adequately capture volatility clustering and sudden political shocks, the analysis shifted to daily data, narrowing the sample window to the period from April 2021 to April 2026. This choice improved the model's ability to capture che determinants of the current economic regime.

The dataset was constructed using a rolling window approach combined with a traditional train-test split:

- A rolling window is a dynamic estimation technique where the model is fitted over a fixed-length data span ($T = 5$ years) that moves forward sequentially through time. Instead of using a static starting point, the estimation window discards the oldest observation as a new one becomes available.

- Within each fixed 5-year window, the data is partitioned into two distinct subsets to prevent overfitting and ensure valid out-of-sample testing:

  1. Training Set (In-Sample): The first 80% of the window—equivalent to exactly 4 years of daily observations—is dedicated to parameter estimation,and diagnostic testing (Ljung-Box and ARCH tests).
  2. Test Set (Out-of-Sample): The remaining 20% of the window—equivalent to 1 year of daily observations—is strictly reserved for forecasting evaluation.

3.2 Methodology

Although model specifications vary across different forecasting weeks, the entire analytical process is governed by a standardized methodological protocol. This approach ensures consistency, replicability, and comparability of results, allowing the impact of specification choices on forecasting performance to be isolated. To do this, we used the function $set.seed(420)$ to ensure exact replicability of the R code for every group member, and consistent data output for all stochastic methods. It is important to note that all outputs provided in this report are produced using the excel data from 30/04/2026. The framework is structured into four operational phases: preprocessing and exploratory analysis, model selection and estimation, forecasting strategy, and out-of-sample validation.

3.2.1 Standard Preprocessing Protocol

Preprocessing follows a rigorous sequence uniformly applied to all time series analyzed:

1. **Import and Conversion:**

   Data are imported from Excel files and converted into time-series objects (zoo/xts classes) to ensure proper handling of temporal indices and frequencies.

2. **Descriptive Analysis :** Key statistical moments (mean, median, variance, skewness, and kurtosis) are computed to characterize the empirical distribution of the series and identify potential fat tails or asymmetries.

3. **Alignment and Cleaning:**

   Multiple series are aligned through inner joins; missing or inconsistent values are removed to avoid estimation distortions.

4. **Sign Inversion:**

   When raw data follow a sign convention opposite to market standards, multiplication by -1 is applied.

5. **Stationarity Testing:**

   The original series is tested using ADF (null hypothesis: unit root) and KPSS (null hypothesis: stationarity). If non-stationarity is detected in levels, first differencing (d = 1) is applied to obtain an integrated I(0) series.

6. **Heteroskedasticity Identification:**

   Analysis of autocorrelation functions (ACF/PACF) on squared residuals reveals volatility clustering. The ARCH-LM test (p-value = 0.000) statistically confirms the presence of ARCH effects, making the inclusion of a GARCH component essential.

7. **Reconstruction to Levels for Validation:**

   Once the model is estimated on the differenced series $\Delta Spread_t$, forecasts and confidence intervals are initially expressed as daily changes. To evaluate performance in economically meaningful terms and compute comparable accuracy metrics against market data, cumulative reconstruction is applied:

$$\widehat{Spread}_t = Spread_{t-1} + \Delta\widehat{Spread}_t.$$

   This step is essential for calculating MAE, RMSE, and empirical coverage on the test set.

Chapter 4

ARMA Model

4.1 Sprecification and Estimation

4.1.1 Preliminary Stationarity Analysis

ARMA model identification begins with visual inspection of autocorrelation functions. The ACF of the original series shows very slow and nearly linear decay, with bars remaining high across many lags. This is the classic signal of a non-stationary series (unit root process), indicating that today's shock has a persistent effect that does not quickly dissipate. The PACF shows a significant spike only at lag 1, suggesting that if an autoregressive component exists, it would likely be AR(1), but non-stationarity dominates the analysis. The original series is therefore not suitable for standard ARMA modeling. Differencing is required.

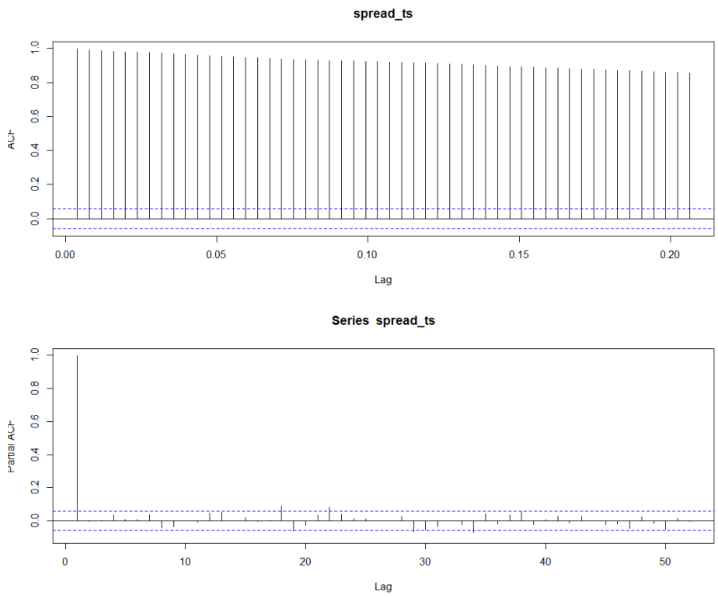

Figure 4.1: ACF and PACF of ARMA model

4.1.2 Formal Stationarity Tests

To rigorously confirm the order of integration, the Augmented Dickey-Fuller (ADF) test is implemented via the ur.df() function from the urca package, allowing finer control than standard implementations. The test is conducted under three configurations:

- type = "none": stationarity around zero
- type = "drift": stationarity around a constant non-zero mean
- type = "trend": stationarity around a deterministic trend

```
================================================
Stationarity test (ADF): Spread BTP/Bund
================================================
   Type ADF_stat Tau_5pct         Result
1  none   -0.362    -1.95 NOT Stationary
2 drift   -1.744    -2.86 NOT Stationary
3 trend   -1.921    -3.41 NOT Stationary
```

Figure 4.2: Stationarity of ARMA model

Decision rule: Reject the null hypothesis of a unit root only if the T statistic is more negative than the 5% critical value.
All the statistics exceed critical values → fail to reject unit root

4.1.3 Differencing and Stationarity Verification

Since the original series is I(1), it must be transformed into an I(0) process by working with daily changes rather than absolute levels.

ADF results on the differenced series:

```
================================================
Stationarity test (ADF): Differenced spread BTP/Bund
================================================
   Type ADF_stat Tau_5pct     Result
1  none  -22.608    -1.95 Stationary
2 drift  -22.597    -2.86 Stationary
3 trend  -22.642    -3.41 Stationary
```

Figure 4.3: Stationarity verification

The differenced series is strongly stationary (I(0)). The transformation is successful, and the series is ready for ARMA modeling.

4.1.4 Autocorrelation Analysis of the Differenced Series

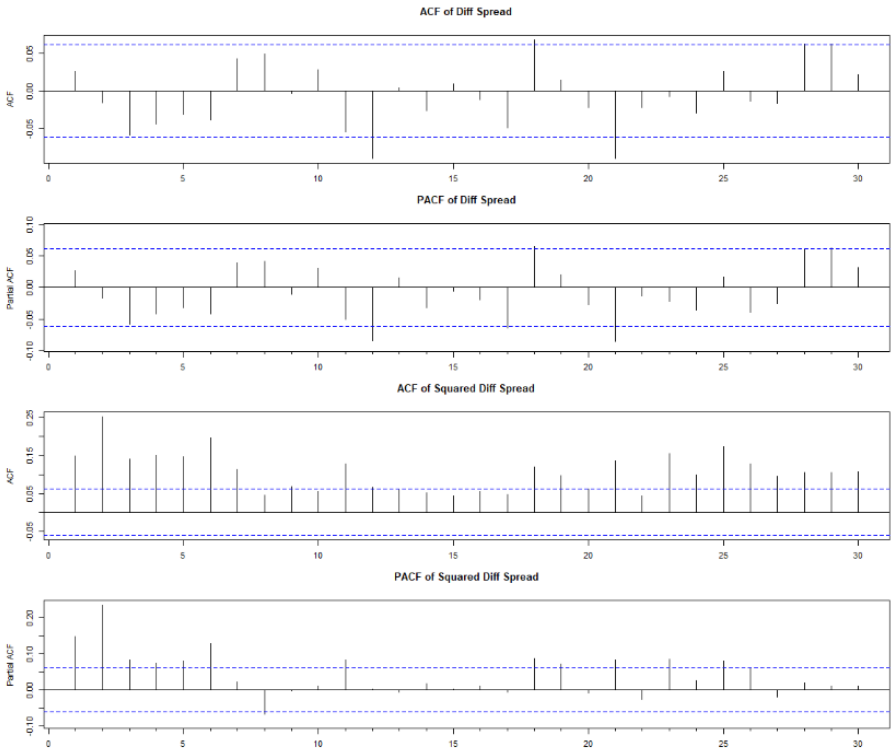

Figure 4.4: Autocorrelation analysis

ACF/PACF of Differenced Spread: No significant lags exceed confidence bands (except one isolated spike at lag 12, ignored to avoid overfitting). Interpretation: The differenced series behaves like White Noise in the mean. No significant linear autocorrelation exists. Implication: The ARMA component will likely be ARMA(0,0).

ACF/PACF of Squared Differenced Spread: Significant spikes appear in early lags. Interpretation: Presence of volatility clustering. Implication: A GARCH model is necessary for variance dynamics.

4.1.5 Preliminary Tests Confirming Structure

To formally validate the graphical observations, we conduct statistical tests:

Test	p-value	H_0	Conclusion
Ljung-Box on Diff Spread	0-0474	No autocorrelation in mean	Reject H_0: autocorrelation detected, need mean modelling
Ljung-Box on Squared Diff Spread	0.0000	No autocorrelation in variance	Reject H_0: strong autocorrelation in squared residuals; GARCH needed
ARCH-LM	0.0000	No ARCH effects	Reject H_0: GARCH needed

4.2 ARIMA Estimation

The *auto.arima* function is used to select the model with minimum AIC.

Although the initial diagnostics—including the visual inspection of the correlogram (ACF/PACF) and the multivariate Ljung-Box test—suggest that the residual series is free from significant serial correlation, the automated auto.arima algorithm selects an ARIMA$(2, 0, 2)$ specification for the conditional mean.. The inclusion of autoregressive (AR) and moving average (MA) terms by the algorithm does not necessarily reflect true linear autocorrelation in the mean. Instead, it is highly likely a result of the strong ARCH effects (autoregressive conditional heteroskedasticity) present in the data. Unmodeled time-varying volatility and clustering can distort standard information criteria (such as AIC and BIC), leading the automated selection process to overfit the conditional mean in an attempt to absorb the non-linear dynamics of the variance. This confirms the necessity of moving beyond standard linear models and explicitly accounting for the volatility process through a multivariate GARCH framework. Compared to a simple white noise stochastic process, it can be noticed by looking both at AIC and BIC and error measures that the models are pretty much similar, model m1 add no improvements to simple white noise.

```
Best ARIMA model:
Series: d_train
ARIMA(2,0,2) with zero mean

Coefficients:
         ar1      ar2      ma1     ma2
      1.5317  -0.9214  -1.5396  0.9025
s.e.  0.0795   0.0560   0.0951  0.0761

sigma^2 = 15.16:  log likelihood = -2831.88
AIC=5673.77   AICc=5673.83   BIC=5698.41
```

```
Series: as.numeric(d_train)
ARIMA(0,0,0) with non-zero mean

Coefficients:
        mean
      0.0028
s.e.  0.1224

sigma^2 = 15.31:  log likelihood = -2838.31
AIC=5680.61   AICc=5680.62   BIC=5690.47
```

Figure 4.5: ARIMA best model

To ensure the statistical validity of the conditional mean, we formally tested the ARMA assumptions through root diagnostics. The results confirm that the process is fully stationary and invertible with no common root, thereby validating the ARIMA$(2, 0, 2)$ specification as structurally sound and parsimonious.

4.2.1 ARIMA Residual Diagnostics

Diagnastic test on the ARMA(2,2) model were performed:

Test	**p-value**	**Conclusion**
Ljung-Box on residual mean	0.5833	ARMA(2,2) captured all linear structure
Ljung-Box on squared residuals	0.0000	Volatility structure remains unexplained
ARCH-LM on residuals	0.0000	ARCH effects confirmed
Jarque-Bera	< 0.05	Residuals are non-normal
KS test (Student-t)	0.9265	Residuals follow Student's t-distribution

As expected there is no linear serial autocorrelation on the residuals series, but there still is conditional heteroskedasticity. In addition, the results show that the residuals are distributed like a Student-t distribution, evidencing the presence of fat tails.

4.3 GARCH Model: Specification and Selection

4.3.1 Motivation for GARCH

Preliminary analysis reveals three fundamental properties:

- Volatility Clustering: periods of high volatility cluster over time
- Fat Tails: empirical distribution exhibits heavier tails than normal
- Persistence: volatility shocks have long-lasting effects

These violate the homoskedasticity assumption of standard ARIMA models, making a GARCH (Generalized Autoregressive Conditional Heteroskedasticity) framework necessary.

4.3.2 Systematic Search for Optimal Model (Grid Search)

To select the optimal specification, we implement a parallel grid search function that systematically tests parameter combinations. Search parameters:

- **ARMA orders :** 0-3 (to avoid overfitting)
- **GARCH orders :** (1,1) fixed (sufficient for most financial series)
- **GARCH Models :** sGARCH (standard), sGARCH-M (ARCH-in-mean)
- **Distributions :** std (Student-t), sstd (Skewed Student-t)

- **Selection criterion :** BIC (penalizes complexity more than AIC

	ar_p <int>	ma_q <int>	garch_p <int>	garch_q <int>	model <chr>	dist <chr>	aic <dbl>	bic <dbl>	likelihood <dbl>
1	0	0	1	1	sGARCH	std	5402.53	5427.17	-2696.27
2	1	0	1	1	sGARCH	std	5403.01	5432.58	-2695.51
3	0	1	1	1	sGARCH	std	5403.04	5432.60	-2695.52
4	0	0	1	1	sGARCH-M	std	5404.49	5434.05	-2696.24
5	0	0	1	1	sGARCH	sstd	5404.52	5434.09	-2696.26
6	1	1	1	1	sGARCH	std	5403.19	5437.69	-2694.60

```
Best ARMA-GARCH model orders:
ARMA:    0 0
GARCH:   1 1
Model:   sGARCH
Distribution:   std
```

Figure 4.6: Best ARMA-GARCH Models

The best model selected by minimum BIC is: ARMA(0, 0) - sGARCH(1, 1) with Student-t distribution.This result shows that once the time-varying volatility is explicitly modeled through a GARCH$(1, 1)$ framework, the optimal specification for the conditional mean reduces to an ARMA$(0, 0)$. This confirms that the higher-order terms initially selected by the automated ARIMA algorithm were driven by the need to absorb unmodeled ARCH effects.

4.3.3 Positivity and Stationarity Conditions

To ensure the mathematical validity of the GARCH model, we verify these three critical conditions:

```
============================================================
GARCH POSITIVITY & STATIONARITY CHECK: ARMA-GARCH (training fit)
============================================================
  omega  = +0.621802  > 0?  YES ✓
  alpha1 = +0.187573  >= 0? YES ✓
  beta1  = +0.781923  >= 0? YES ✓

  Persistence (sum α + sum β) = 0.969496  < 1? YES ✓ (covariance stationary)
  Unconditional variance       = 20.384260
  Unconditional vol (daily)    = 4.5149 bps
  --> All positivity conditions SATISFIED ✓
============================================================
```

Figure 4.7: Conditions to satisfy for the model to work

Economic interpretation:

- $\omega(Omega)$: baseline volatility floor always present on the market
- $\alpha_1(Alpha)$: immediate reactivity (19% of shock transmitted instantly)
- $\beta_1(Beta)$: persistence (79% carried into next day)
- $\alpha_1 + \beta_1 = 0.96949 < 1$: highly persistent but stationary

4.3.4 Full GARCH Diagnostics

To validate the model, we conduct a battery of tests on standardized residuals.

Test	**Results**	**Conclusion**
Weighted Ljung-Box Test on Standardized Residuals	high p-values: 0.71, 0.89, 0.80	There is no residual autocorrelation. The model has captured the entire linear structure.
Weighted Ljung-Box Test on Standardized Squared Residuals	high p-values: 0.3497, 0.71, 0.76	There is no residual volatility clustering. The GARCH model captures the variance structure.
Weighted ARCH LM Tests	high p-values: 0.45, 0.78, 0.91	No remaining ARCH effects
Sign Bias Test	high p-values: 0.75, 0.59, 0.99, 0.80	There is no asymmetry, the symmetric sGARCH model is adequate
Adjusted Pearson Goodness-of-Fit Test	high p-values: 0.51, 0.60, 0.63, 0.69	The residuals fit well to the Student-t distribution

4.3.5 Rolling Validation Strategy

The out-of-sample valuation is conducted using a rolling estimation procedure with a 252-day moving window (approximately one trading year), a choice consistent with risk management practices to capture a complete cycle of market conditions. As the window advances, the oldest observation is dropped and a new one is added ensuring that the estimated parameters remain highly responsive to the most recent financial conditions without being contaminated by past structural breaks. The model parameters is re-estimated every 10 observations to update the parameters to new volatility regimes, balancing accuracy and computational efficiency. For each step, a one-day-ahead forecast ($n.ahead = 1$) is generated with confidence intervals based on the estimated t-Student distribution. This procedure simulates a real-world operating environment with daily risk monitoring, avoiding look-ahead bias and ensuring that performance reflects the model's ability to adapt to new information.

Chapter 5

ARMA-GARCH Validation

5.1 Out-of-Sample Validation with Rolling Window

The GARCH model was validated using a 252-day rolling window (approximately one trading year), with one-step-ahead forecasts ($n.ahead = 1$). Parameters were re-estimated every 10 iterations to balance adaptability and computational efficiency.

Metric	**Value**	**Interpretation**
MAE	1.54 bps	Average forecasting error of 1.54 bps
RMSE	2.21 bps	Penalizes larger errors, consistent with MAE
MAPE	Inf	Not usable due to negative spreads
Dir. Acc.	53.33%	Slightly above random chance, useful but weak for directional trading

5.2 Distribution Evaluation and Interval Coverage

Beyond point forecasts, the core utility of the GARCH framework lies in its ability to generate dynamic, risk-managed confidence intervals that adapt to time-varying volatility.Distribution evaluation measures whether realized values fall within model-predicted confidence intervals. A well-calibrated model should show empirical coverage close to theoretical coverage.

- 90% Confidence Interval: The empirical coverage stands at 94.90%, exceeding the theoretical target by nearly 5%. This indicates that the model is conservative at this

threshold, generating slightly wider intervals than necessary and capturing a higher proportion of realized spread values.

- 98% Confidence Interval: At the extreme tails, the empirical coverage matches the theoretical expectation almost perfectly at 98.82%.

Confidence Level	**Theoretical Coverage**	**Empirical Coverage**	**Interpretation**
90%	90%	94.90%	Slightly conservative (intervals somewhat wider)
98%	98%	98.82%	Excellent tail calibration

Chapter 6

VAR Model

6.1 Data Setup

To construct a robust Vector Autoregression (VAR) framework for forecasting purposes, the primary dataset—initially composed solely of the BTP-BUND spread—was augmented with a selection of exogenous financial time series. These variables, sourced from the Refinitiv Eikon platform, were selected based on established economic theory regarding the determinants of sovereign risk and Eurozone monetary dynamics. To preliminary select the variable that could be used to construct a VAR model useful to the objective, a Granger Causality Test was performed to identify the specific variables providing statistically significant information for predicting future values of the spread. Prior to testing, all series underwent a transformation process to ensure stationarity, thereby avoiding the issue of spurious regressions. The test was structured to evaluate the null hypothesis (H_0) that the lagged values of a candidate variable do not improve the forecast of the spread. The significance threshold was set at 10% ($\alpha = 0.10$). Based on the empirical results, the null hypothesis was successfully rejected for three specific variables. These indicators demonstrate a predictive lead over the BTP-BUND spread and were consequently incorporated into the final VAR specification:

- **EUROIS3M (3-Month Euribor-OIS Spread):** This indicator measures the differential between the 3-month Euribor and the 3-month Overnight Index Swap rate. It serves as a primary barometer for banking system stress; an expansion in this differential signals rising counterparty risk .

- **ESDE (Spain-Germany Sovereign Spread):** This variable represents the yield differential between Spanish 10-year government bonds and the German Bund. It captures the "peripheral contagion" effect; since international investors often perceive Italy and Spain as having correlated risk profiles, stress in the Spanish bond market frequently acts as a leading indicator for movements in the BTP-BUND spread.

- **IEDE (Ireland-Germany Sovereign Spread):** This variable represents the yield differential between Irish 10-year government bonds and the German Bund. Although Ireland's fiscal dynamics diverge from Italy's, this spread serves as an important indicator of systemic risk within the Eurozone periphery (as Italy and Spain, Ireland is a member of the so called P.I.I.G.S.). Movements in the Irish spread signal broader shifts in investor sentiment toward "non-core" European debt, providing early warning signals for asset reallocation.

Consistent with the univariate case, the dataset covers a 5-year period. A rolling window approach was implemented, where the first 80% of the sample (corresponding to 4 years of observations) was utilized as the training set, while the remaining 20% served as the out-of-sample test set. Missing values within the dataset were addressed using linear interpolation via the na.approx() function. To ensure data integrity, any initial missing observations—specifically those arising from the first-difference transformation required to achieve stationarity—were systematically excluded from the sample. Finally, to compute the Orthogonalized Impulse Response Functions (OIRF), the variables within the VAR system were ordered as follows: EUROIS3M, ESDE, IEDE, and Spread. This specific ordering reflects a structural identification scheme based on a Cholesky decomposition, placing the most exogenous market variables first and the BTP-BUND spread last, as it is assumed to be the most endogenous variable reacting contemporaneously to shocks in the other series.

6.2 Methodology

In standard Vector Autoregression (VAR) analyses, the optimal lag length is typically determined using information criteria that evaluate the system as a whole. However, because the primary objective of this research is specifically the forecasting of the BTP-BUND spread rather than the simultaneous estimation of all variables in the system, a targeted selection approach was adopted. Instead of optimizing the global fit of the VAR, the lag length was selected to maximize the Adjusted R^2 of the individual spread equation, that is 5 lags.Although the resulting Adjusted R^2 remains relatively low (below 0.03). After excluding the coefficients were not statistically significant at the 10% level ($\alpha = 0.10$), the final specification of the spread equation was determined as follows:

```
Estimation results for equation spread:
=======================================
spread = eurois3m.l1 + IEDE.l1 + eurois3m.l2 + IEDE.l3 +
eurois3m.l4 + spread.l4 + ESDE.l5 + spread.l5

            Estimate Std. Error t value Pr(>|t|)
eurois3m.l1 -9.68452    4.66177  -2.077 0.038015 *
IEDE.l1      0.16340    0.07335   2.228 0.026126 *
eurois3m.l2 18.29642    5.08072   3.601 0.000332 ***
IEDE.l3     -0.22416    0.07335  -3.056 0.002301 **
eurois3m.l4 -7.35478    4.30285  -1.709 0.087707 .
spread.l4   -0.05277    0.03109  -1.697 0.089963 .
ESDE.l5      0.21752    0.10938   1.989 0.047019 *
spread.l5   -0.11474    0.05645  -2.033 0.042348 *
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1


Residual standard error: 3.863 on 1006 degrees of freedom
Multiple R-Squared: 0.03539,    Adjusted R-squared: 0.02772
F-statistic: 4.613 on 8 and 1006 DF,  p-value: 1.512e-05
```

Figure 6.1: VAR spread sprecification with statistically significant coefficients

6.2.1 MQ Test

To ensure the statistical validity of the estimated Vector Autoregression (VAR) model and to verify that the residuals behaved as white noise, a series of diagnostic tests were performed on the residual series. Overall, the results confirm that the residuals are free from significant serial correlation and can be statistically treated as white noise in the mean, validating the lag structure of the VAR.

```
Ljung-Box Statistics:
          m       Q(m)      df    p-value
 [1,]   1.00      5.29  -10.00     1.00
 [2,]   2.00     17.56    6.00     1.00
 [3,]   3.00     25.70   22.00     0.26
 [4,]   4.00     44.01   38.00     0.23
 [5,]   5.00     65.82   54.00     0.13
 [6,]   6.00     90.29   70.00     0.05
 [7,]   7.00    104.31   86.00     0.09
 [8,]   8.00    126.34  102.00     0.05
 [9,]   9.00    147.20  118.00     0.04
[10,]  10.00    158.04  134.00     0.08
```

Figure 6.2: MQ test of the spread model

At the first two horizons, the parameter correction results in negative or severely restricted degrees of freedom ($df = -10.00$ at lag 1 and $df = 6.00$ at lag 2), rendering the corresponding p-values of 1.00 uninformative. However, from lag 3 onward, the test functions properly. At lag 3, the test statistic $Q(3) = 25.70$ generates a p-value of 0.26, and at lag 4, $Q(4) = 44.01$ results in a p-value of 0.23. An isolated, marginal rejection is observed only at lag 9 ($Q(9) = 147.20$, p-value $= 0.04$) . However, since this minor

deviation occurs at a distant lag while all critical short-term horizons strictly support the null hypothesis, it does not provide sufficient evidence to reject the overall adequacy of the model. By the final lag horizon ($m = 10$), the p-value returns to 0.08, confirming that the null hypothesis of no serial correlation is not rejected.

6.2.2 Distribution

To determine the appropriate distribution for the innovation process, the empirical properties of the VAR residuals were formally tested. Initially, a Jarque-Bera (JB) test was conducted to assess the null hypothesis of multivariate normality. The test overwhelmingly rejected the null hypothesis of normality (p-value < 0.01), driven by significant excess kurtosis ($K > 3$) and negative skewness ($S \neq 0$) typical of financial time series during periods of market stress.

```
$spread

        JB-Test (univariate)

data:  Residual of spread equation
Chi-squared = 429.38, df = 2, p-value < 2.2e-16


$JB

        JB-Test (multivariate)

data:  Residuals of VAR object model_s
Chi-squared = 22486, df = 8, p-value < 2.2e-16
```

Figure 6.3: Jarque Bera test on the spread model

To further identify the most appropriate distribution, the residuals of the single equations were formally compared under a Normal and a Student's t alternative using the Akaike and Bayesian Information Criteria (AIC and BIC). Focusing on the BTP-BUND spread equation, the Student's t distribution achieved an AIC of 5491.67 and a BIC of 5506.44, which are substantially lower than the Normal distribution alternatives (AIC = 5614.11, BIC = 5623.95). This empirical evidence confirms the presence of heavy tails in the innovation process, justifying the use of a Student's t distribution to prevent estimation bias.

6.2.3 Granger Test

To validate the predictive power of the selected covariates within the restricted Vector Autoregression (VAR) framework, a directed Granger causality test was conducted with the specific objective of assessing whether the candidate variables contain predictive information for the BTP-BUND spread

Granger test on restricted and non restricted models:

```
$Granger

        Granger causality H0: eurois3m ESDE IEDE do not
        Granger-cause spread

data:  VAR object model_s
F-Test = 1.3105, df1 = 15, df2 = 3972, p-value = 0.1861


$Instant

        H0: No instantaneous causality between: eurois3m ESDE
IEDE
        and spread

data:  VAR object model_s
Chi-squared = 412.31, df = 3, p-value < 2.2e-16


$Granger

        Granger causality H0: eurois3m ESDE IEDE do not
        Granger-cause spread

data:  VAR object model
F-Test = 2.4216, df1 = 15, df2 = 3972, p-value = 0.001644


$Instant

        H0: No instantaneous causality between: eurois3m ESDE
IEDE
        and spread

data:  VAR object model
Chi-squared = 413.33, df = 3, p-value < 2.2e-16
```

Figure 6.4: Granger test output

The Granger causality test for the restricted VAR model (model_s) yielded a p-value of 0.2088, failing to reject the null hypothesis at standard significance levels. We observed that the causality requirements are satisfied at the two extremes of the restriction spectrum: the unrestricted level (threshold = 0) and a highly parsimonious level (threshold >= 2.3). In the former case, the inclusion of all lags provides a comprehensive filter that captures the full systemic dynamics. In the latter, the test is passed by isolating only the most dominant predictors, which reduces the degrees of freedom and inflates the F-statistic. Despite the apparent statistical success of the model at thresholds > =2.3, that configuration was rejected due to the re-emergence of residual autocorrelation in the multivariate Ljung-Box (MQ) diagnostics. Consequently, the 1.64 threshold was maintained. Although it presents a more challenging environment for the Granger test, it is the only specification that prevents both the overfitting of the full model and the diagnostic invalidity of the over-restricted model, ensuring a robust foundation for the subsequent volatility analysis. Furthermore, it is important to note that for both specifications the null hypothesis of no instantaneous causality is rejected (p-value < 0.01). This confirms that while the lagged transmission channels are highly sensitive to the restriction threshold, the contemporaneous interactions

among variables remain strong.

6.2.4 Impulse Response Function

The dynamic responsiveness of the BTP-BUND spread was analyzed through Orthogonalized Impulse Response Functions (OIRF), employing a structural identification scheme based on a Cholesky decomposition. The variables were entered into the VAR system in the following order: EUROIS3M, ESDE, IEDE, and Spread. This specific ordering is theoretically grounded in established economic literature, as it treats interest rate differentials and peripheral sovereign risk indicators as relatively more exogenous, thereby allowing them to exert a contemporaneous impact on the BTP-BUND spread while the spread reacts to all other innovations with a lag. Based on the empirical significance of the lagged coefficients (at the 95% confidence level), the transmission dynamics reveal several key insights:

- **Contemporaneous and Short-Term Transmission from Spain (ESDE):** A primary finding is the immediate sensitivity of the BTP-BUND spread to shocks originating from the Spanish sovereign market. The yield differential for Spain (ESDE) demonstrates statistical significance at both lag 0 and lag 3.
- **Monetary Policy and Interbank Stress Expectations (EUROIS3M):** The Euribor-OIS spread (EUROIS3M) demonstrates a significant impact on the BTP-BUND spread at lag 2.
- **Lagged Transmission from the Irish Periphery (IEDE):** Shocks from the Irish sovereign spread (IEDE) exhibit a slightly delayed propagation mechanism, achieving statistical significance at lag 3.

6.2.5 FEVD

To analyze the structural composition of the forecasting uncertainty of the BTP-BUND spread, the Forecast Error Variance Decomposition (FEVD) was computed over a 20-day horizon using a Cholesky identification scheme. The empirical results reveal that the uncertainty of the Italian spread is overwhelmingly driven by regional spillovers rather than domestic innovations. This confirms that the BTP-BUND spread is highly exposed to market sentiment regarding the broader Eurozone periphery. The idiosyncratic component—the extent to which the spread explains its own variance—starts at 31.89% at $h = 1$. This indicates that roughly one-third of the unexpected movements in the Italian spread are due to purely domestic factors or country-specific structural shocks.

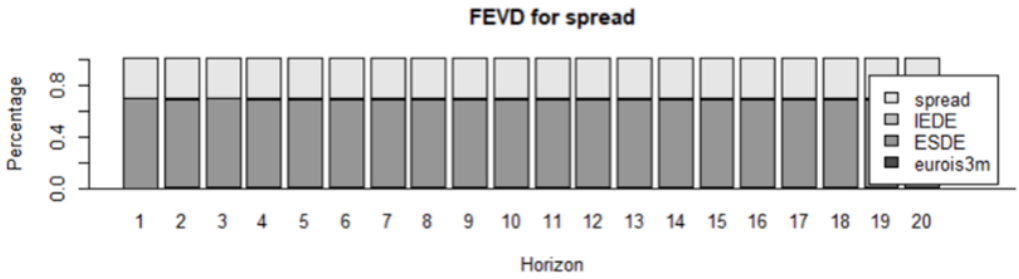

Figure 6.5: FEVD output of the spread model

6.2.6 ARCH-LM Test

Finally an ARCH-LM test was performed on the residuals of the restricted VAR. The test detected the presence of significant conditional heteroskedasticity (ARCH effects), a common feature in high-frequency financial time series. While these ARCH effects do not alter the point estimates of the standard FEVD—which relies on the unconditional variance-covariance matrix— they imply that the true variance proportions are time-varying, consequently the output of FEVD should be interpreted as a long-run sample average. The univariate ARCH-LM test on the residuals of the BTP-BUND spread equation strongly rejects the null hypothesis of no ARCH effects. Similarly, the multivariate ARCH test applied to the entire VAR system rejects the null hypothesis of homoskedasticity. Because both diagnostic tests confirm that the constant variance assumption of the standard VAR is violated, it is statistically necessary to explicitly model the time-varying variance-covariance structure of the residuals. To address this conditional heteroskedasticity and capture the dynamic co-movements among the variables, the analysis proceeds by implementing a Dynamic Conditional Correlation (DCC-GARCH) model.

Chapter 7

DCC Model, Validation, and Forecasting

7.1 Introduction

This section presents the estimation, diagnostic validation, and out-of-sample forecasting evaluation of the VAR-DCC model applied to the BTP–BUND spread and its associated interest rate and the Ireland and Spain spread variables. The primary objective is to assess whether a multivariate volatility modelling improves our ability to forecast the spread between BTP and BUND. Following the Vector Autoregression (VAR) estimation described in the previous section, we extract the VAR residuals to fit a Dynamic Conditional Correlation (DCC) GARCH model, which relaxes the restrictive assumption of constant cross-series correlations that would otherwise be imposed on the system.

7.2 Motivation: Conditional Heteroskedasticity in VAR Residuals

The theoretical justification for augmenting the VAR model with a GARCH-DCC part relies on the presence of time-varying volatility in the VAR residuals. A constant-covariance assumption would be inappropriate if the residuals exhibit ARCH effects, leading to inefficient parameter estimates and unreliable inference. We formally test this using the multivariate ARCH Lagrange Multiplier (LM) test applied to the residuals of the estimated VAR model (model_s).

The test has a Chi-squared statistic of 1905.5 on 500 degrees of freedom, with an associated p-value below 2.2×10^{-16}, providing overwhelming evidence against the null hypothesis of no conditional heteroskedasticity. This result confirms that the residual covariance matrix is not constant over time and that volatility modeling is both statistically necessary and economically meaningful in this context; financial series are known to exhibit

volatility clustering.

7.3 DCC Model Estimation

7.3.1 Distributional Choice

A preliminary step in the DCC estimation is the selection of the innovation distribution. Motivated by a multivariate Jarque–Bera test on model_s residuals, which strongly rejects normality, and by information criteria computed over competing distributional assumptions, we evaluate both Gaussian and Student-t specifications for each univariate margin. For each of the four series, we estimate sGARCH(1,1) models under both normal and t-distributed innovations and compare them via Akaike (AIC) and Bayesian (BIC) Information Criteria. The results are presented in the following table.

	spread — AIC	spread — BIC	eurois3m — AIC	eurois3m — BIC	ESDE — AIC	ESDE — BIC	IEDE — AIC	IEDE — BIC
t-distribution	2849.05	2863.81	2892.28	2907.04	2802.26	2817.02	2568.72	2583.48
Normal distribution	2870.98	2880.83	3021.68	3031.52	2880.57	2890.42	2782.98	2792.48

Table 7.1: Information criteria for sGARCH(1,1) under t and normal innovations across the four series. Lower values indicate preferred specifications.

Across all the series and both criteria, the Student-t specification is uniformly preferred, in some cases by a substantial margin. This strong and consistent preference supports the use of fat-tailed innovations, which is consistent with the well-documented leptokurtosis of financial time series. Accordingly, we proceed with t-distributed sGARCH(1,1) models for each univariate component.

7.3.2 DCC Parameter Estimates

The multivariate DCC(1,1) model is estimated using the two-step Engle (Engle, 2002) procedure: univariate GARCH parameters are estimated in the first step, and the DCC correlation parameters are estimated in the second step conditional on the standardised residuals. All models assume a multivariate Student-t distribution. The optimal parameter estimates are reported below.

```
*---------------------------------*
*          DCC GARCH Fit          *
*---------------------------------*

Distribution         :  mvt
Model                :  DCC(1,1)
No. Parameters       :  25
[VAR GARCH DCC UncQ] : [0+16+3+6]
No. Series           :  4
No. Obs.             :  1014
Log-Likelihood       :  -3052.958
Av.Log-Likelihood    :  -3.01

Optimal Parameters
------------------------------------
                    Estimate  Std. Error  t value Pr(>|t|)
[eurois3m].omega    0.000006    0.000009  0.73363 0.463176
[eurois3m].alpha1   0.212375    0.069278  3.06554 0.002173
[eurois3m].beta1    0.786625    0.095680  8.22142 0.000000
[eurois3m].shape    5.057222    0.821622  6.15517 0.000000
[ESDE].omega        0.271928    0.171711  1.58364 0.113276
[ESDE].alpha1       0.117947    0.052652  2.24014 0.025082
[ESDE].beta1        0.813695    0.089237  9.11831 0.000000
[ESDE].shape        5.313457    0.777737  6.83195 0.000000
[IEDE].omega        0.449240    0.505583  0.88856 0.374240
[IEDE].alpha1       0.374170    0.244146  1.53257 0.125382
[IEDE].beta1        0.588684    0.309400  1.90266 0.057085
[IEDE].shape        3.234645    0.307116 10.53231 0.000000
[spread].omega      0.595285    0.300262  1.98256 0.047417
[spread].alpha1     0.163449    0.054657  2.99043 0.002786
[spread].beta1      0.803872    0.064352 12.49188 0.000000
[spread].shape      7.800683    1.593581  4.89507 0.000001
[Joint]dcca1        0.028414    0.007875  3.60817 0.000308
[Joint]dccb1        0.873403    0.033490 26.07945 0.000000
[Joint]mshape       4.898892    0.220830 22.18403 0.000000

Information Criteria
---------------------

Akaike       6.0709
Bayes        6.1923
Shibata      6.0697
Hannan-Quinn 6.1170
```

The univariate GARCH estimates reveal important differences in volatility dynamics across the four series. The spread exhibits high volatility persistence, with the sum of ARCH ($\alpha_1 = 0.163$) and GARCH ($\beta_1 = 0.804$ coefficients approaching unity (0.967), suggesting near-integrated volatility dynamics — shocks to spread volatility dissipate very slowly. The degrees of freedom parameter (shape = 7.801) indicates moderately heavy tails. The eurois3m series presents a larger ARCH coefficient $\alpha_1 = 0.212$, indicating that short-term OIS volatility responds more sharply to new information, with overall persistence also high $\alpha_1 + \beta_1 = 0.999$, approaching an IGARCH process. The ESDE spread shows a similar pattern, with high persistence $\alpha_1 + \beta_1 = 0.932$ and a significant ARCH effect $\alpha_1 = 0.118$, p = 0.025). The IEDE spread stands out with the largest ARCH coefficient ($\alpha_1 = 0.374$), indicating the strongest short-term reaction to new shocks, though with lower overall persistence ($\alpha_1 + \beta_1 = 0.963$) and a notably lower shape parameter (3.235), suggesting considerably heavier tails compared to the other series. At the DCC level, the parameters dcca1 α and dccb1 β govern the dynamics of the conditional correlation matrix. The estimated dccb1 = 0.873 is highly significant ($p < 0.001$), confirming that

correlations are highly persistent and evolve smoothly over time. The dcca1 = 0.028 is also statistically significant (p = 0.000), suggesting that recent cross-products of lagged shocks do contribute to updating conditional correlations, though the effect is small relative to the persistence term. The joint shape parameter (mshape = 4.899, $p < 0.001$) confirms the appropriateness of the multivariate Student-t distribution, capturing the heavy-tailed joint behavior of the series.

7.3.3 Dynamic Conditional Correlations

Figure 7.1 displays the estimated pairwise dynamic conditional correlations over the train period. Three pairs are shown: spread vs. ESDE, spread vs. eurois3m, and spread vs. IEDE. The spread:ESDE pair (red) exhibits the highest and most stable correlation throughout the sample, consistently clustering near 0.8, with only 2 notable downward spikes around mid-2023 and mid-2024. This reflects the very close co-movement between the BTP-Bund spread and the Spanish-German sovereign spread. The spread:IEDE pair (blue) shows a moderate and relatively stable correlation, fluctuating in the 0.35–0.55 range for most of the sample, with occasional transient spikes toward 0.60, suggesting a meaningful but less tight relationship between the BTP–Bund and the Irish–German spread. The spread:eurois3m pair (green) fluctuates around zero throughout the entire period, with frequent negative excursions reaching as low as -0.15, highlighting the episodic and nearly orthogonal relationship between the BTP–Bund spread and short-term money market conditions. These time-varying correlation properties — particularly the sharp temporary drops in the spread:ESDE pair and the sign-switching behavior of spread:eurois3m — would not be captured by a constant-correlation model such as CCC-GARCH, or worse, by a plain VAR model. The DCC specification thus adds meaningful information about the evolving co-movement structure that would otherwise be lost.

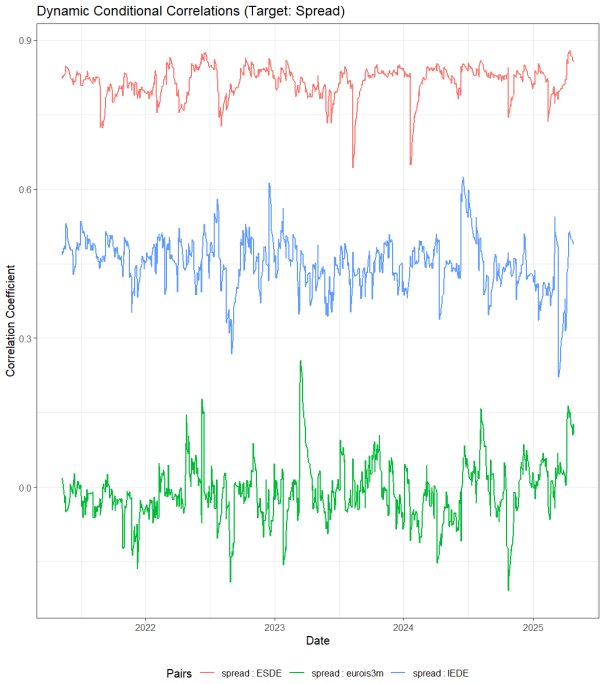

Figure 7.1: Dynamic Conditional Correlations — spread:ESDE (red), spread:eurois3m (blue), spread:IEDE (green), 2021–2025

7.4 Model Diagnostics

Following estimation, we conduct a set of diagnostic tests on the DCC standardised residuals to assess whether the model has adequately captured the conditional mean, variance, and correlation dynamics of the system. The MarchTest is applied to the squared standardised residuals to detect remaining ARCH effects after the VAR-DCC filtering. Three out of four test variants confirm adequate model fit: the LM test (Q(m) = 8.769, p = 0.554), the multivariate Portmanteau test on squared series (Q_k(m) = 151.18, p = 0.679), and the robust variant (154.02, p = 0.618) all fail to reject the null of no remaining ARCH effects, indicating that the DCC specification successfully absorbs the volatility clustering in the residuals. The sole exception is the rank-based test (statistic = 33.78, p = 0.0002), which rejects the null. The sole exception is the rank-based test (statistic = 33.78, p = 0.0002), which rejects the null. This test is a non-parametric alternative that is specifically sensitive to nonlinear dependence and outliers in the tails of the distribution, rather than to standard second-moment dynamics. Its rejection here is likely attributable to the presence of extreme observations — consistent with the heavy-tailed behavior already documented by the low shape parameter of the IEDE series (shape = 3.23) and the joint mshape $\bar{4}$.90. Furthermore, the rank-based test is picking up tail dependence that is not necessarily a sign of model misspecification, but rather a reflection of the inherently fat-tailed nature of the data that even a multivariate Student-t distribution can only partially accommodate.

Q(m) of squared series (LM test): statistic = 8.769, p-value = 0.554 Rank-based test: statistic = 33.78, p-value = 0.0002 Q_k(m) of squared series: statistic = 151.18, p-value = 0.679 Robust test (5%): statistic = 154.02, p-value = 0.618

Overall, the parametric diagnostics confirm that the model is correctly specified in terms of volatility dynamics, and we proceed to the out-of-sample forecasting evaluation with this caveat in mind.

7.5 Model Validation: Rolling-Window Forecasting

7.5.1 Forecasting Methodology

To evaluate the predictive performance of the VAR-DCC model, we adopt a rolling-window out-of-sample forecasting exercise with a one-step-ahead horizon (h = 1). The window size is fixed to match the estimation sample used for model_s, ensuring comparability across iterations. At each step, a VAR(5) model is re-estimated and the same zero restrictions identified during the initial model reduction are re-imposed, preserving the approved specification structure while allowing coefficients to adapt to new information. The DCC-GARCH model is also re-estimated at each window on the updated VAR residuals, so both the conditional mean and the conditional covariance matrix are refreshed at every iteration. Forecasted differences are then converted back to the original level scale by accumulating the one-step-ahead increments. The 95% forecast intervals are derived from the DCC-forecasted covariance matrix, scaled by the quantile of a Student-t distribution with degrees of freedom equal to the estimated joint shape parameter. Three metrics are computed to evaluate forecasting accuracy: Root Mean Squared Error (RMSE): measures average forecast deviation from realised values. Directional Accuracy (Acc. Dir.): the fraction of forecasts correctly predicting the sign of the next-period change. Confidence Interval Coverage (CI Coverage, 95%): the proportion of realised values falling within the estimated 95% forecast intervals.

7.5.2 Forecast Performance

The forecasting results for all three endogenous variables are presented in the table below:

Variable	**RMSE**	**MAE**	**MAPE**	**Acc. Dir.**	**CI Cov. 0.95**
eurois3m	2.3636	1.6488	2.0414	47.2441	98.8235
ESDE	0.0251	0.0171	116.5261	49.2126	96.8627
IEDE	6.8187	1.5264	5.2564	53.5433	3.5294
spread	1.2382	0.7800	2.5035	51.5748	98.4314

The spread RMSE (1.238) is substantially lower than the Random Walk benchmark (2.200), indicating that the VAR-DCC model produces meaningfully more accurate point

forecasts for the primary variable of interest. Confidence interval coverage is above the nominal 95% level for three out of four series — spread (98.43%), eurois3m (98.82%), and ESDE (96.86%) — suggesting that the DCC-based forecast intervals are slightly conservative. The notable exception is IEDE, whose CI coverage collapses to just 3.53%, indicating a severe underestimation of forecast uncertainty for the Irish–German spread, likely related to its high ARCH coefficient and heavier tails documented in the estimation stage. Directional accuracy hovers near 50% for spread (51.57%), eurois3m (47.24%), and ESDE (49.21%), equivalent to a coin toss, confirming that the model provides no reliable signal for predicting the direction of movement. The IEDE series reaches the highest directional accuracy (53.54%), though this remains modest. Overall, the model's main contribution lies in level forecast accuracy for the spread relative to the Random Walk, rather than in directional or interval forecasting.

7.5.3 Benchmark Comparison: Random Walk

To assess whether the VAR-DCC model delivers any improvement over a naive forecast, we benchmark against a Random Walk (RW) model defined as: Spread_t+1 = Spread_t The Random Walk achieves an RMSE of 2.2004 basis points on the spread series. The VAR-DCC model records an RMSE of 1.238, a reduction of approximately 44This indicates that the VAR-DCC model does provide a meaningful improvement in point forecast accuracy over the naive benchmark, contrary to what a purely visual inspection might suggest. However, directional accuracy for the spread remains at 51.57%, barely above chance, meaning that while the model produces closer forecasts in terms of magnitude, it offers no reliable signal about the direction of the next movement. This result is visible in Figure 3, where the VAR-DCC and Random Walk forecasts follow a broadly similar path, both tracking the actual spread closely within the wide 95% confidence bands — a consequence of the high volatility persistence documented in the GARCH estimation stage. In summary, the VAR-DCC model adds value in terms of level accuracy but not in terms of directional predictability, which is the more practically relevant criterion for a trading or hedging context.

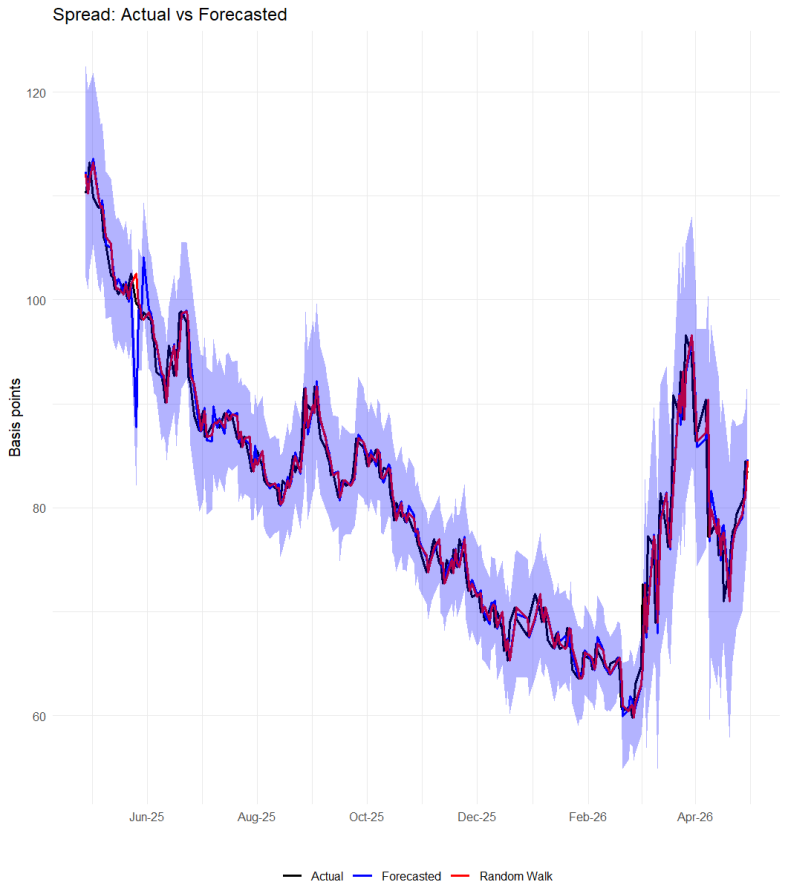

Figure 7.2: Spread — Actual vs. VAR-DCC Forecast vs. Random Walk, with 95% confidence bands (basis points, Jul 2025 – May 2026

7.6 Discussion and Conclusions

The VAR-DCC framework estimated in this work satisfies all key statistical requirements. Conditional heteroskedasticity is clearly present in the VAR residuals, Student-t innovations are strongly preferred over the normal distribution across all series, the DCC parameters are statistically significant and economically interpretable, and the diagnostic tests confirm adequate model fit — with the sole exception of the rank-based test, whose rejection is attributable to tail dependence rather than to a structural misspecification. The estimated dynamic conditional correlations reveal meaningful time-variation in the co-movements between the BTP–Bund spread and the other variables, most notably the very high and persistent correlation with the ESDE spread and the near-zero, sign-switching relationship with eurois3m — features that a constant-correlation specification such as CCC-GARCH would entirely miss. Despite this in-sample validity, the out-of-sample forecasting evaluation yields a more nuanced picture. On one hand, the VAR-DCC model does outperform the Random Walk in terms of RMSE, achieving a 44% reduction in point forecast error for the spread. On the other hand, directional accuracy hovers around 50%

for all series, indicating that the model provides no reliable signal about the direction of the next movement — the criterion that is arguably most relevant from a practical standpoint. The IEDE series additionally presents a near-total collapse in confidence interval coverage (3.53%), suggesting that the model severely underestimates forecast uncertainty for that series, likely due to its heavy-tailed and highly reactive volatility dynamics. Two explanations are plausible for these limitations. First, the system may be missing important exogenous drivers of sovereign spreads — such as ECB policy surprises, fiscal announcements, or market sentiment indicators — that are not spanned by the three endogenous variables included in the model. Second, the one-step-ahead horizon may be too short for the VAR structure to provide a meaningful advantage over a naive benchmark, as the spread behaves close to a random walk at daily frequency. The DCC component, while statistically well-specified and informative for correlation analysis and conditional covariance forecasting, does not translate into improved directional predictability for the spread.